Ryan Vickramasinghe

Software Engineer

Seattle, WA / ryanvickr@gmail.com / 650-213-7527

Professional Summary

Software Engineer with 5+ years of experience specializing in large-scale, data-intensive distributed systems. Genuinely enjoys learning, and building reliable solutions to problems with elegant, scalable code.

- Strong proficiency in C++ and distributed systems.
- Good communicator, problem-solver, and self-motivated.
- Thrives in both team and self-directed environments.
- History of designing new systems as well as improving existing ones.

Employment History

AUG, 2022 – PRESENT

SEATTLE, WA

Software Engineer (L5), Google

BigQuery Team

Working on the BigQuery core runtime team, focused on Parquet performance enhancements across the stack.

- Contributed to design and implementation of a project to migrate our entire query engine to a fully vectorized solution, speeding up query times by an average of 3x. See blog post on BigQuery advanced runtime.
- Proposed, designed, and implemented a full rewrite of the Parquet I/O layer in order to further improve performance of the query engine, resulting in a 20% reduction in I/O wait-time and a massive reduction in code complexity, using multi-threading and caching.
- Found bottlenecks in, and completely redesigned a field matching algorithm resulting in up to 100x speed enhancement for wide tables, drawing in several large customers.
- Designed and implemented various storage-level optimization techniques (e.g. filter pushdown, dictionary folding, expansion pruning).
- Technologies Used: C++, Rust, Python, Arrow, Parquet, Iceberg

Streaming Engines Team

Worked on Google's streaming engine, a very large-scale distributed streaming analytics platform written in C++. The streaming engine was originally created for, and is used by, Search, Ads, YouTube, and more. It is now also offered as a Cloud product (Dataflow, BigQuery Continuous Query).

- Researched, designed, and implemented a large-scale horizontal and vertical auto-tuning system to optimize cost and performance (https://cloud.google.com/dataflow/docs/guides/right-fitting).
- Improved 90th percentile latency of a critical service by 25x by implementing caching.
- Decreased system failure recovery time from 30s to 5s by reducing thread blocking and and adding ref counting.
- Designed and implemented a significant reworking of key components to support streaming SQL (through BigQuery).
- Completed a "20% project" to enhance Youtube's video engine for live streams
- Top-4 code contributor in a 500+ person org.
- Technologies Used: C++, Python, Kubernetes, GCP

MAY, 2021 – AUG, 2021

VANCOUVER, BC

Software Engineer Intern, Amazon

Developed a benchmarking platform for Datapath (Amazon shopping's core infrastructure).

- Increased efficiency through task automation, freeing 100s of engineering hours annually
- Fully redesigned and refactored code to improve and restore website functionality and appearance
- Wrote technical documents and manuals for new software being developed
- Technologies Used: Java, Javascript (React), and AWS (Lambda, DynamoDB, Step Functions, SQS)

MAY, 2020 – OCT, 2020
TORONTO, ON

Software Engineer Intern, Royal Bank Of Canada

Involved in design and implementation of an enterprise-wide data catalogue which aggregates and presents available data using C++ 11 and Java 8.

- Provided leadership as a subject matter expert on graph database management/architecture
- Collaborated with business-users and architects to understand business requirements and translate them into technical solutions
- Technologies Used: Java, C++, Cassandra, NoSQL, MySQL

MAY, 2018 – AUG, 2019
TORONTO, ON

Software Engineer Intern, Royal Bank Of Canada

Designed and implemented a system which reconciles various risk ledgers using graph technology, processing millions of data points using C++ and Java.

- Researched and designed a data model which improved performance by over 60%
- Created an application which allowed users to graphically build complex graph queries to retrieve data from the system
- Evaluating hardware-software requirements, and providing recommendations
- Installed, tuned, and maintained a NoSQL database cluster in a RHEL (Linux) environment
- Technologies Used: Java, C++, Cassandra, NoSQL, MySQL

Technologies

C++, Java, Rust, Python, VHDL, Javascript, Typescript, HTML/CSS, Angular, React, Kubernetes, Docker, GCP, AWS, UNIX/Linux, NoSQL, SQL, Tableau, Performance Tuning

Education

SEP, 2016 – MAY, 2022
LONDON, ON

Bachelor of Computer Engineering, University of Western Ontario

Extracurricular Activities: Intramural Sports Captain, Western SAE Formula One

Links

Github: github.com

LinkedIn: linkedin.com